Field Service Management - Project Workflow & Architecture Guide

1. Overall Architecture

```
React Frontend
      |
HTTP (REST API)
      |
ASP.NET Core Web API
      |
Controllers
      |
Services (later)
      |
Entity Framework Core
      |
SQL Server
```

2. Recommended Backend Folder Structure

```
FieldServiceManagement.API
■
■■■ Controllers
■■■ Models
■■■ DTOs
■■■ Data
■   ■■■ AppDbContext.cs
■■■ Interfaces
■■■ Services
■■■ Helpers
■■■ Migrations
■■■ Properties
■■■ Program.cs
■■■ appsettings.json
```

3. Frontend Roles

```
Customer
- Register/Login
- Browse Services
- Create Service Request
- Track Request
- View Invoice
- Pay Invoice
- Notifications

Technician
- Dashboard
- Assigned Jobs
- Accept Job
- Start Travel
- Start Work
- Submit Work Report

Manager
- Dashboard
- Review Requests
- Assign Technician
- Review Work Report
- Generate Invoice
- Close Request

Owner
- Full Access
- Manage Users
- Manage Services
- View Reports
- System Settings
```

4. Complete Business Workflow

```
1. Customer Login
2. Customer Creates Service Request
3. Manager Reviews Request
4. Manager Assigns Technician
5. Technician Accepts Job
6. Technician Travels
7. Technician Starts Work
8. Technician Submits Work Report
9. Manager Reviews Report
10. Manager Approves Report
11. Manager Generates Invoice
12. Customer Pays Invoice
13. Manager Closes Request
```

5. Database Flow

```
Users
 ↓
Customer Address
 ↓
Services
 ↓
Service Requests
 ↓
Work Reports
 ↓
Invoices
 ↓
Invoice Items
 ↓
Notifications
 ↓
Status History
```

6. Backend Development Order

```
Phase 1
- SQL Server
- EF Core
- AppDbContext
- User Model
- Migration
- Test API

Phase 2
- Authentication
- Users
- Customer Addresses

Phase 3
- Services
- Technicians
- Service Requests

Phase 4
- Work Reports
- Invoice
- Notifications

Phase 5
- Dashboard
- Settings
- JWT
- Authorization
```

7. API Development Workflow

```
Model
 ↓
DbContext
 ↓
```

```
Migration
 ↓
Database Update
 ↓
Controller
 ↓
Scalar Testing
 ↓
React Integration
```

8. Request Lifecycle

```
React Form
 ↓
POST API
 ↓
Controller
 ↓
DbContext
 ↓
SQL Server
 ↓
Response
 ↓
React UI Update
```

9. Dashboard Data

```
Customer Dashboard
- Active Requests
- Pending Invoice
- Completed Jobs

Technician Dashboard
- Assigned Jobs
- Today's Schedule
- Completed Jobs

Manager Dashboard
- Pending Requests
- Technicians
- Work Reports
- Invoices

Owner Dashboard
- Revenue
- Active Jobs
- Employees
- Services
```

10. Final Build Roadmap

```
1. Foundation
2. Authentication
3. User Management
4. Customer Module
5. Technician Module
6. Service Module
7. Service Request Module
8. Work Report Module
9. Invoice Module
10. Notification Module
11. Dashboard Module
12. Frontend Integration
13. Testing
14. Deployment
```

11. Interview Explanation

```
Explain:
- Why ASP.NET Core?
- Why Code First?
- Why EF Core?
```

- Why SQL Server?
- Why React?
- Explain complete request flow.
- Explain database relationships.
- Explain authentication and authorization.
- Explain service request lifecycle.